

1. (100 pts) Using partial-fraction-expansions, convert the following Laplace-domain signals to the time domain. Make sure to show your steps and reasoning. Make sure to also verify your answers using a software tool, e.g., Mathematica/MATLAB/Wolfram Alpha/etc.

a) $X(s) = \frac{5s+12}{(s+2)(s+5)}$

b) $X(s) = \frac{s^3+10s^2+7}{(s+2)(s+3)^3}$

c) $X(s) = \frac{s(s+4)e^{-s}}{(s-3)(s^2-4)}$

d) $X(s) = \frac{4s+11}{(s-5)(s+9)}$

e) $X(s) = \frac{2s^2-11s+2}{(s-1)^2(s-2)}$

f) $X(s) = \frac{6s^3+45s^2+14s+12}{(s+4)^2(s+5)(s+6)}$

g) $X(s) = \frac{9s^2+2s+7}{(s^2+2s+25)^2}$

h) $X(s) = \frac{(s^2+2s+5)(s-2)}{(s-1)(s^2+4s+20)}$

i) $X(s) = \frac{6s^4+10s+7}{(s^2+2s+5)}$

j) $X(s) = \frac{3s^3+7s^2+19s+21}{(s^2+5s+36)^2}$